MODERN PHYSICS

Mock Test 2

Time: 50 minutes Maximum marks: 30

Instructions. This paper has 30 multiple-choice questions. Each question has exactly one correct option and carries 1 mark. Mark your answers on the OMR sheet.

Data. $h = 6.63 \times 10^{-34}\,\text{J s}$, $\hbar = 1.055 \times 10^{-34}\,\text{J s}$, $c = 3 \times 10^{8}\,\text{m/s}$, $hc = 1240\,\text{eV nm}$, $1\,\text{eV} = 1.6 \times 10^{-19}\,\text{J}$, $e = 1.6 \times 10^{-19}\,\text{C}$, $m_e = 9.11 \times 10^{-31}\,\text{kg}$, $\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F/m}$, $R = 1.097 \times 10^{7}\,\text{m}^{-1}$, $a_0 = 0.53\,\text{Å}$.

Q1. In a plane electromagnetic wave in vacuum, the electric field at a certain point and instant is 60 V/m. The total (electric plus magnetic) energy density at that point and instant is

(A) 1.6×10^{-8} J/m^3 (B) 3.2×10^{-8} J/m^3 (C) 6.4×10^{-8} J/m^3 (D) 1.3×10^{-7} J/m^3

Q2. Maxwell added a new term to Ampère's law. Its physical content is that

(A) a changing magnetic flux induces an emf in a closed loop

(B) isolated magnetic poles do not exist

(C) a changing electric flux gives rise to a magnetic field, even in empty space

(D) a stationary charge produces an electric field

Q3. In a double-slit experiment the slit separation is 0.20 mm and the screen is 1.0 m away. Adjacent bright fringes are 3.0 mm apart. The wavelength of the light is

(A) 300 nm (B) 450 nm (C) 500 nm (D) 600 nm

Q4. A laser of wavelength 500 nm emits a beam of power 2.0 mW. The number of photons emitted per second is about

(A) 5×10^{15} (B) 5×10^{18} (C) 5×10^{12} (D) 1×10^{16}

Q5. In a photoelectric experiment, the stopping potential V_s is plotted against the frequency ν of the incident light for a given metal, giving a straight line. The slope of the line and its intercept on the frequency axis are respectively

(A) h/e and the threshold frequency ν_0

(B) e/h and the threshold frequency ν_0

(C) h/e and the work function ϕ in volts

(D) ϕ/e and h/e

Q6. A metal has a work function of 2.0 eV. The longest wavelength of light that can eject photoelectrons from it is

(A) 310 nm (B) 620 nm (C) 1240 nm (D) 2480 nm

Q7. Electrons are emitted thermionically from a heated filament with negligible speed and are accelerated through 100 V. Their speed on arrival is about

(A) 5.9×10^{5} m/s (B) 4.2×10^{6} m/s (C) 5.9×10^{6} m/s (D) 5.9×10^{7} m/s

Q8. Photons pass through a single slit of width a and the sideways spread of their momentum, Δp_y, is measured. If the slit width is halved, Δp_y approximately

(A) halves (B) stays the same (C) doubles (D) quadruples

Q9. The momentum of a photon of wavelength 600 nm is

(A) 1.1×10^{-36} kg m/s (B) 3.3×10^{-19} kg m/s (C) 1.1×10^{-25} kg m/s (D) 1.1×10^{-27} kg m/s

Q10. A ball of mass 0.20 kg moves at 10 m/s. Its de Broglie wavelength is

(A) 3.3×10^{-34} m (B) 6.6×10^{-34} m (C) 1.3×10^{-33} m (D) 3.3×10^{-30} m

Q11. In one dimension, the probability density $|\psi|^2$ of a particle near the point x_0 is 2.0×10^{9} m^{-1}. The probability of finding the particle in a small interval of width 0.50 Å around x_0 is about

(A) 1% (B) 10% (C) 20% (D) 100%

Q12. At a point on a screen the wavefunctions Ψ_1 and Ψ_2 of two separate contributions are both real and each equals 0.30. For the superposition $\Psi = \Psi_1 + \Psi_2$, the probability density $|\Psi|^2$ at that point is

(A) 0.09 (B) 0.18 (C) 0.36 (D) 0.60

Q13. The momentum operator is $\hat{p}_x = -i\hbar\,\dfrac{d}{dx}$. When it acts on $\psi(x) = e^{ikx}$, the eigenvalue obtained is

(A) $\hbar k$ (B) $-\hbar k$ (C) $\hbar/k$ (D) $i\hbar k$

Q14. For a particle in a one-dimensional infinite well of width L (walls at $x = 0$ and $x = L$), in the state $n = 3$, the expectation values $\langle x\rangle$ and $\langle p_x\rangle$ are

(A) $L/6$ and 0 (B) $L/2$ and $3\pi\hbar/L$

(C) $L/3$ and 0 (D) $L/2$ and 0

Q15. An electron meets a potential barrier whose height exceeds its energy by 1.0 eV. The penetration depth, $\delta = \hbar/\sqrt{2m(U-E)}$, is about

(A) 0.2 Å (B) 2 Å (C) 20 Å (D) 200 Å

Q16. In alpha decay, the alpha particle escapes from the nucleus even though its energy is lower than the height of the Coulomb barrier at the nuclear surface. The explanation is that

(A) the alpha particle gains energy from thermal vibrations of the nucleus until it exceeds the barrier

(B) the Coulomb barrier at the nuclear radius is lower than the energy of the alpha particle

(C) the strong force is repulsive at the nuclear surface and pushes the alpha particle out

(D) the wavefunction of the alpha particle is nonzero beyond the barrier, so there is a small probability of tunnelling through it

Q17. A diatomic molecule vibrates as a harmonic oscillator with force constant $k = 500$ N/m and reduced mass $\mu = 1.0 \times 10^{-26}$ kg. The frequency of the photon emitted in a transition $n \to n-1$ is about

(A) 2.2×10^{14} Hz (B) 3.6×10^{13} Hz (C) 7.1×10^{13} Hz (D) 1.4×10^{14} Hz

Q18. Which of the following can **not** be a value of L_z for an electron with orbital quantum number $l = 2$?

(A) 0 (B) $\hbar$ (C) $1.5\hbar$ (D) $-2\hbar$

Q19. The magnitude of the spin angular momentum of an electron is

(A) $\hbar/2$ (B) $\frac{\sqrt{3}}{2}\hbar$ (C) $\hbar$ (D) $\frac{3}{4}\hbar$

Q20. Which of the following particles obeys the Pauli exclusion principle?

(A) a photon (spin 1)

(B) a Higgs particle (spin 0)

(C) an electron (spin $\frac{1}{2}$)

(D) a hypothetical particle of spin 2

Q21. The spin-orbit energy is proportional to $\vec{L}\cdot\vec{S}$ with a positive constant. Comparing the $2p_{3/2}$ and $2p_{1/2}$ states,

(A) in $2p_{3/2}$ the vectors $\vec{L}$ and $\vec{S}$ are more aligned, $\vec{L}\cdot\vec{S} > 0$, and the energy is higher

(B) in $2p_{3/2}$ the vectors $\vec{L}$ and $\vec{S}$ are opposed and the energy is lower

(C) both states have the same $\vec{L}\cdot\vec{S}$, so there is no splitting

(D) in $2p_{1/2}$ the vectors $\vec{L}$ and $\vec{S}$ are more aligned and the energy is higher

Q22. In the characteristic X-ray spectrum of an element, K_α arises from the $L \to K$ transition and K_β from the $M \to K$ transition. Compared with K_α, the K_β line has

(A) lower photon energy and longer wavelength

(B) lower photon energy and shorter wavelength

(C) the same photon energy and wavelength

(D) higher photon energy and shorter wavelength

Q23. The H_α line ($n = 3 \to 2$) of deuterium lies at a slightly shorter wavelength than that of hydrogen. This is because

(A) the neutron adds extra positive charge to the deuterium nucleus

(B) the electron in deuterium has a larger rest mass

(C) the nucleus is stationary in deuterium but moves in hydrogen

(D) the larger nuclear mass makes the reduced mass of the electron-nucleus system slightly larger, so all energy gaps are slightly larger

Q24. Ignoring spin, the number of distinct hydrogen states (l, m_l) with principal quantum number $n = 3$ is

(A) 3 (B) 6 (C) 9 (D) 18

Q25. J.J. Thomson found that cathode rays had the same charge-to-mass ratio whatever the cathode material or residual gas. He concluded that

(A) the particles in cathode rays are a common constituent of all atoms

(B) atoms are indivisible

(C) the positive charge of an atom is concentrated in a small nucleus

(D) the mass of an atom is entirely due to its electrons

Q26. The number of alpha particles scattered into an angle θ by a thin foil varies as $1/\sin^4(\theta/2)$. For the same foil and beam, the ratio of the number scattered at 60° to the number scattered at 180° is

(A) 2 (B) 4 (C) 16 (D) 8

Q27. Which of the following is **not** a postulate of Bohr's model of the hydrogen atom?

(A) An electron moves only in certain allowed circular orbits, each with a definite energy.

(B) An electron in an allowed orbit radiates at its orbital frequency and so spirals into the nucleus.

(C) Radiation is emitted only when the electron jumps to a lower orbit, with $h\nu = E_n - E_{n'}$.

(D) An atom can absorb a photon of energy $h\nu = E_{n'} - E_n$ and move to a higher orbit.

Q28. The radius of the $n = 3$ Bohr orbit of hydrogen is about

(A) 4.8 Å (B) 1.6 Å (C) 3.2 Å (D) 14 Å

Q29. The longest wavelength in the Paschen series of hydrogen ($n \to 3$) is about

(A) 820 nm (B) 1280 nm (C) 4050 nm (D) 1875 nm

Q30. The hydrogen transition $2 \to 1$ has wavelength 121.6 nm and the transition $3 \to 2$ has wavelength 656.3 nm. Using the Rydberg-Ritz combination principle, the wavelength of the $3 \to 1$ transition is about

(A) 103 nm (B) 535 nm (C) 778 nm (D) 92 nm

End of question paper

Answer Key and Solutions

1	2	3	4	5	6	7	8	9	10
B	C	D	A	A	B	C	C	D	A

11	12	13	14	15	16	17	18	19	20
B	C	A	D	B	D	B	C	B	C

21	22	23	24	25	26	27	28	29	30
A	D	D	C	A	C	B	A	D	A

Q1. (B) *Energy in EM waves, Lecture 1-2*

In vacuum $B = E/c$, which makes the magnetic and electric energy densities equal at every point and instant. Hence $u = 2u_E = \varepsilon_0 E^2 = (8.85 \times 10^{-12})(60)^2 = 3.2 \times 10^{-8}$ J/m^3. Option A is the electric part alone.

Q2. (C) *Maxwell's equations, Lecture 1-2*

Faraday's law already says a changing magnetic flux gives an electric field. Maxwell's new term is the symmetric statement: a changing electric flux (displacement current) gives a magnetic field. The two together allow a self-sustaining wave in empty space.

Q3. (D) *Double-slit interference, Lecture 1-2*

Fringe spacing $\Delta y = \lambda D/d$, so $\lambda = \dfrac{d\,\Delta y}{D} = \dfrac{(0.20 \times 10^{-3})(3.0 \times 10^{-3})}{1.0} = 6.0 \times 10^{-7}$ m $= 600$ nm.

Q4. (A) *Photons and intensity, Lecture 1-2*

Photon energy $= 1240/500 = 2.48$ eV $= 3.97 \times 10^{-19}$ J. Rate $= P/E_\gamma = \dfrac{2.0 \times 10^{-3}}{3.97 \times 10^{-19}} \approx 5 \times 10^{15}$ s^{-1}.

Q5. (A) *Stopping potential graph, Lecture 1-2*

$eV_s = h\nu - \phi$ gives $V_s = \dfrac{h}{e}\,\nu - \dfrac{\phi}{e}$. The slope is h/e (the same for every metal) and $V_s = 0$ at $\nu_0 = \phi/h$, the threshold frequency. The intercept on the V_s axis is $-\phi/e$.

Q6. (B) *Threshold wavelength, Lecture 1-2*

$\lambda_0 = \dfrac{hc}{\phi} = \dfrac{1240 \text{ eV nm}}{2.0 \text{ eV}} = 620$ nm. Any longer wavelength has photons below 2.0 eV and ejects nothing.

Q7. (C) *Electron acceleration, Lecture 3*

Energy gained from the field equals the kinetic energy: $\frac{1}{2}mv^2 = eV$, so $v = \sqrt{\dfrac{2eV}{m}} = \sqrt{\dfrac{2(1.6 \times 10^{-19})(100)}{9.11 \times 10^{-31}}} = 5.9 \times 10^6$ m/s. This is about 2% of c, so the non-relativistic formula is adequate. Option B results from dropping the factor 2.

Q8. (C) *Single-slit uncertainty, Lecture 4*

The position uncertainty in y is about the slit width, $\Delta y \approx a$, so $\Delta p_y \gtrsim h/a$. Halving a doubles Δp_y. This matches the wider diffraction pattern from a narrower slit.

Q9. (D) *Photon momentum, Lecture 4*

$p = \dfrac{h}{\lambda} = \dfrac{6.63 \times 10^{-34}}{6.0 \times 10^{-7}} = 1.1 \times 10^{-27}$ kg m/s. Option B is the photon energy in joules (hc/λ), not a momentum.

Q10. (A) *de Broglie wavelength of a macroscopic object, Lecture 4*

$\lambda = \dfrac{h}{mv} = \dfrac{6.63 \times 10^{-34}}{0.20 \times 10} = 3.3 \times 10^{-34}$ m. This is far smaller than any structure it could meet, so wave behaviour is never seen.

Q11. (B) *Probability density, Lecture 5*

For a small interval, $P \approx |\psi|^2\,\Delta x = (2.0 \times 10^9)(5.0 \times 10^{-11}) = 0.10 = 10\%$.

Q12. (C) *Superposition, Lecture 5*

Amplitudes add, not probabilities: $\Psi = 0.30 + 0.30 = 0.60$, so $|\Psi|^2 = 0.36$. Adding the separate probabilities would give $0.09 + 0.09 = 0.18$, which misses the interference enhancement.

Q13. (A) *Operators and eigenvalues, Lecture 5*

$\hat{p}_x e^{ikx} = -i\hbar\,(ik)\,e^{ikx} = \hbar k\, e^{ikx}$. The eigenvalue is $\hbar k$, the momentum of a wave of wavenumber k.

Q14. (D) *Expectation values in a box, Chapter 5*

For every n, $|\psi_n|^2 \propto \sin^2(n\pi x/L)$ is symmetric about $L/2$, so $\langle x\rangle = L/2$. The state is a standing wave made of equal amounts of $+p$ and $-p$, so $\langle p_x\rangle = 0$.

Q15. (B) *Barrier penetration, Chapter 5*

$2m(U-E) = 2(9.11\times10^{-31})(1.6\times10^{-19}) = 2.9\times10^{-49}$, whose square root is 5.4×10^{-25}. Then $\delta = \dfrac{1.055\times10^{-34}}{5.4\times10^{-25}} \approx 1.95\times10^{-10}$ m ≈ 2 Å.

Q16. (D) *Tunnelling, Chapter 5*

Classically the particle cannot cross a barrier higher than its energy. Quantum mechanically the wavefunction decays inside the barrier but does not vanish, leaving a small probability of finding the alpha particle outside. Option B is false because the barrier is higher than the alpha energy.

Q17. (B) *Harmonic oscillator, Chapter 5*

The classical frequency is $\nu = \dfrac{1}{2\pi}\sqrt{\dfrac{k}{\mu}} = \dfrac{1}{2\pi}\sqrt{5\times10^{28}} = \dfrac{2.24\times10^{14}}{2\pi} = 3.6\times10^{13}$ Hz. Adjacent levels are $h\nu$ apart, so the emitted photon has this frequency. Option A is the angular frequency ω in rad/s.

Q18. (C) *Space quantization, Lecture 5*

$L_z = m_l\hbar$ with $m_l = -2,-1,0,1,2$. The value $1.5\hbar$ would need a non-integer m_l.

Q19. (B) *Spin, Lecture 5*

$S = \hbar\sqrt{s(s+1)} = \hbar\sqrt{3/4} = \dfrac{\sqrt{3}}{2}\hbar \approx 0.87\hbar$. The value $\hbar/2$ is the largest component S_z along an axis.

Q20. (C) *Fermions and bosons, Lecture 6*

Particles with half-integer spin are fermions, and their joint wavefunction is antisymmetric under exchange, $\Psi(1,2) = -\Psi(2,1)$. Two identical fermions in the same state would give $\Psi = 0$, so the Pauli principle follows. Integer-spin particles (photon, Higgs, spin 2) are bosons.

Q21. (A) *Spin-orbit coupling, Lecture 6*

$j = l + \frac{1}{2}$ corresponds to $\vec{L}$ and $\vec{S}$ more nearly parallel, so $\vec{L}\cdot\vec{S} > 0$ and, with a positive coefficient, the energy is higher. For $j = l - \frac{1}{2}$ the vectors are more opposed, $\vec{L}\cdot\vec{S} < 0$, and the energy is lower.

Q22. (D) *K_β versus K_α, Lecture 6*

The M shell lies higher than the L shell, so the $M \to K$ energy gap is larger than the $L \to K$ gap. A larger photon energy means a shorter wavelength.

Q23. (D) *Reduced mass and isotope shift, Chapter 4*

The Bohr energies scale with the reduced mass $\mu = \dfrac{mM}{m+M}$. A heavier nucleus gives μ slightly closer to m, so μ increases, the levels become slightly deeper and every gap slightly larger. Larger energy means shorter wavelength.

Q24. (C) *Degeneracy, Chapter 6*

For $n = 3$: $l = 0$ gives 1 state, $l = 1$ gives 3, $l = 2$ gives 5. Total $= 1 + 3 + 5 = 9 = n^2$. Including spin would double this to 18.

Q25. (A) *Thomson's experiment, Chapter 4*

A particle with the same e/m appearing for every cathode material and every gas shows that it is a universal constituent of matter. It does not by itself show a nucleus (that came later, from Rutherford's experiment) and atoms are not entirely made of electrons.

Q26. (C) *Rutherford scattering formula, Chapter 4*

$N \propto 1/\sin^4(\theta/2)$. At 60°: $\sin 30° = \frac{1}{2}$, so $1/\sin^4 = 16$. At 180°: $\sin 90° = 1$, so $1/\sin^4 = 1$. The ratio is 16.

Q27. (B) *Bohr postulates, Chapter 4*

Radiation by an orbiting electron with spiralling collapse is the classical prediction, which Bohr's model removes. Bohr postulates that an electron in an allowed orbit does not radiate, and that radiation is emitted or absorbed only in transitions, with $h\nu$ equal to the energy difference.

Q28. (A) *Bohr radius, Chapter 4*

$r_n = n^2 a_0$, so $r_3 = 9a_0 = 9(0.53) = 4.8\,\text{Å}$. Options B, C and D correspond to $3a_0$, $6a_0$ and $27a_0$.

Q29. (D) *Paschen series, Chapter 4*

$\dfrac{1}{\lambda} = R\left(\dfrac{1}{3^2} - \dfrac{1}{4^2}\right) = \dfrac{7R}{144}$, so $\lambda = \dfrac{144}{7(1.097 \times 10^7)} = 1.875 \times 10^{-6}$ m $= 1875$ nm. This is the smallest gap in the series. Option A is the series limit $9/R$, B is $5 \to 3$, and C is the first line of the Brackett series ($5 \to 4$).

Q30. (A) *Rydberg-Ritz combination principle, Chapter 4*

Wavenumbers add for chained transitions: $\dfrac{1}{\lambda_{31}} = \dfrac{1}{\lambda_{21}} + \dfrac{1}{\lambda_{32}} = \dfrac{1}{121.6} + \dfrac{1}{656.3} = 9.75 \times 10^{-3}$ nm^{-1}, so $\lambda_{31} \approx 102.6$ nm ≈ 103 nm. Check: $\dfrac{1}{\lambda} = R\left(1 - \dfrac{1}{9}\right) = \dfrac{8R}{9}$ gives 102.6 nm. Options B and C come from subtracting and adding wavelengths directly.